

ANVESHAK ((M5G7))

AI-Powered Indian Standards & Procurement Intelligence Platform

Smart India Hackathon 2026 (Internal Evaluation) • Complete Presentation, Architecture & Defense Dossier

Engineered for GFR 2017 Rule 144 Compliance, Bureau of Indian Standards (BIS), and Gazette Quality Control Orders (QCOs)

CRITICAL BENCHMARKS TO MEMORIZE FOR JUDGES:

150 L+ CRORE

Annual Public Procurement
20-25% of India's GDP (Rule 144)

21,000+ STANDARDS

Bureau of Indian Standards
Across 15 Engineering Divisions

700+ QCOs

Statutory Ministry Orders
Mandatory BIS ISI & MeitY CRS

0 - 100 SCORE

Procurement Readiness Index
Multi-Pillar Verification Metric

01

Problem Statement: The Compliance Crisis in Indian Procurement

Public procurement in India involves massive capital outlays across Central Ministries (MoRTH, MoHUA, MeitY), State PWDs, Defence, Railways, and PSUs (NTPC, BHEL, IOCL). By law (GFR 2017 Rule 144), all public tenders MUST adhere to Indian Standards, and products covered under Quality Control Orders (QCOs) MUST carry certified ISI/CRS marks.

However, procuring officers operate under severe structural handicaps:

- **The Manual Search Nightmare:** Over 21,000 standards across 15 engineering divisions make it impossible for officers to manually identify all identify all required primary, testing, and allied safety standards.
- **Prevalence of Obsolete Standards:** Tenders routinely cite superseded editions (e.g. citing IS 8112 for 43-grade cement or IS 269:1989), triggering contractor disputes, rejected consignments, and court arbitration.
- **Violations of Mandatory QCOs:** Over 700+ product categories now have legally enforceable Gazette Quality Control Orders. Specifying non-certified goods is an explicit statutory violation.
- **Failure of Traditional Keyword Search:** Keyword search engines fail because procurement specifications are unstructured technical descriptions (ratings, voltages, IP protection, testing protocols) that require semantic reasoning and normative dependency traversal.

Winning Soundbite for Judges

"Respected Judges, public procurement errors cost India thousands of crores in litigation, delays, and sub-standard infrastructure. Anveshak converts this manual compliance burden into automated, explainable procurement intelligence in seconds."

02

Master 2-Minute Elevator Pitch (Word-For-Word Presentation Script)

"Good morning Respected Judges. Indian public procurement accounts for over 150 Lakh Crore every year. By the Rules, every single rupee spent on public goods must strictly comply with Indian Standards, and hundreds of products are now under mandatory Gazette Quality Control Orders.

Yet today, procuring officers across ministries and PSUs manually search through 21,000+ BIS standards. As a result, tenders frequently cite obsolete editions, omit mandatory safety testing, and risk costly contractor disputes.

We built ANVESHAK — India's first AI-Powered Indian Standards & Procurement Intelligence Platform.

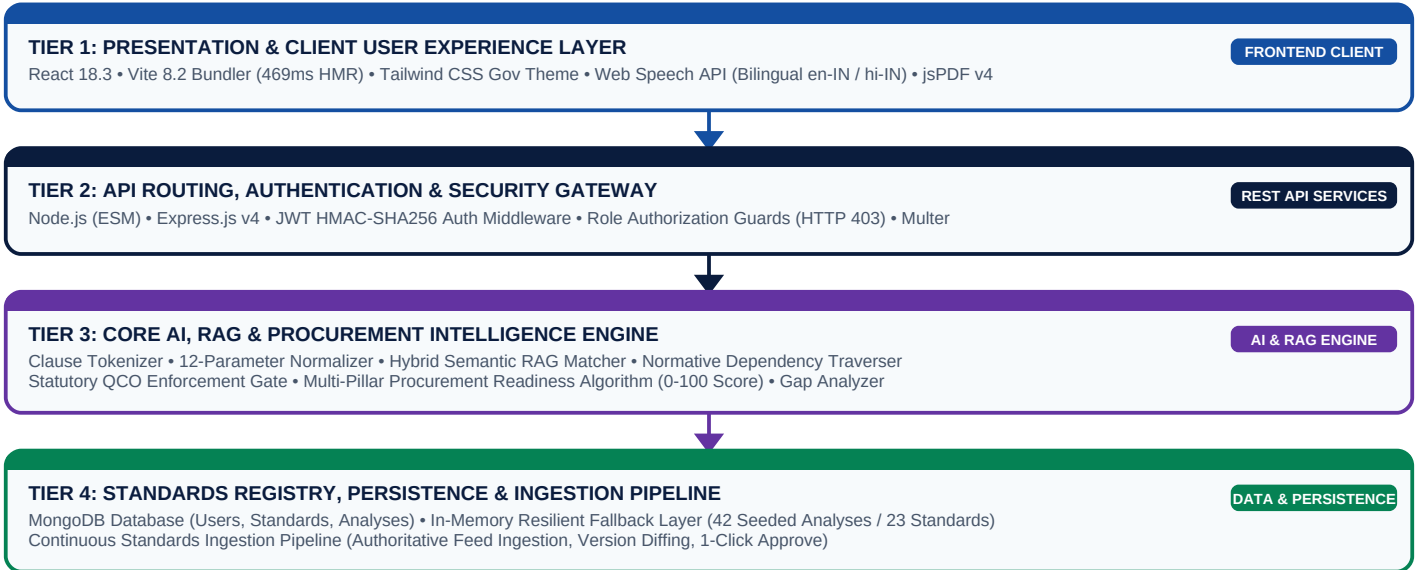
With Anveshak, an officer can upload any multi-page tender document, BOQ schedule, or speak a specification in English or Hindi. In under 3 seconds, our explainable RAG engine extracts critical technical parameters, identifies applicable Indian Standards, verifies mandatory Quality Control Orders, detects missing safety testing clauses, and calculates a Procurement Readiness Score.

Crucially, Anveshak is explainable — every recommendation cites verbatim text evidence from the tender. And unlike static catalogs, our continuous Standards Ingestion Pipeline automatically detects new standards, revisions, and amendments without requiring system rebuilds. Thank you."

03

Complete Visual System Architecture Diagram

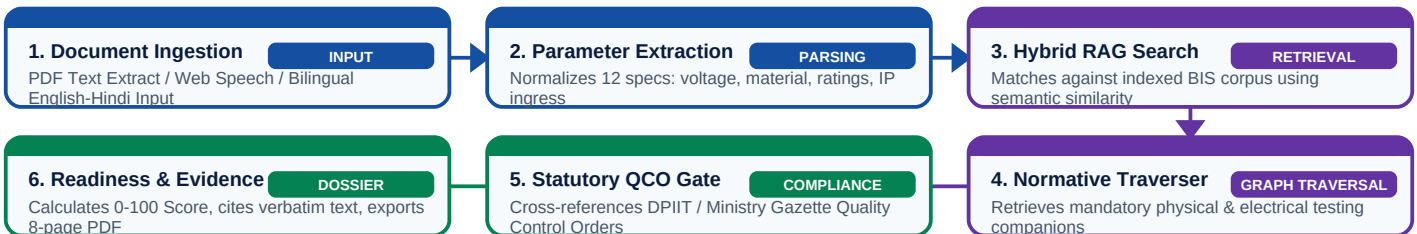
Anveshak implements a modern, resilient 4-tier microservices architecture engineered for high concurrency, zero-downtime offline presentation, and sub-second serverless response times:



04

AI Decision Flowchart: From Tender Upload to Compliance Dossier

The diagram below illustrates the exact deterministic-AI hybrid decision sequence executed when a procurement specification is analyzed:



Key Concept: Hybrid Neural-Symbolic Architecture

Notice that Large Language Models (LLMs) are used strictly for document parsing and semantic intent mapping. All standards matching, testing companions, and QCO deterministically against verified, indexed records. This mathematical guarantee prevents hallucinations.

5-Minute Live Demo Walkthrough (What To Show vs. What To Say)

Follow this exact minute-by-minute cheat-sheet during your live presentation. One team member handles the mouse and screen while the other delivers the voice track:

TIME	ACTION ON SCREEN (WHAT TO CLICK)	EXACT WORDS TO SAY TO THE JUDGES	KEY EMPHASIS
0:00 - 0:45	Open Login Page. Click "Use Demo Account" for Rajesh Kumar (Procurement Officer, CPWD).	"Notice our 1-click evaluation personas. Logging in generates a real HMAC-SHA256 JWT with server-verified role claims."	Server-Enforced RBAC
0:45 - 1:45	Arrive at Officer Dashboard (12 civil/electrical records). Click "New Analysis". Click Voice Input (toggle English/Hindi). Select "100W Outdoor LED Street Light". Click "Analyze Specification".	"Officers can upload multi-page PDFs, type, or speak bilingual specifications. In under 3 seconds, Anveshak normalizes 12 parameters and matches the BIS corpus."	Bilingual Voice & Parameter Extraction
1:45 - 2:45	Result renders: Score 92/100. Click IS 10322; show verbatim quotation. Scroll to Certification Intelligence (BIS ISI Mark Scheme-I, DPIIT QCO). Highlight Gap Analysis: Missing 10kV surge protection.	"Anveshak is fully explainable — it cites the exact phrase from the tender. It also caught a critical compliance gap: missing 10kV surge protection required by IS 16107."	Explainable Evidence & Statutory Gap Detection
2:45 - 3:30	Click "Download PDF Report". Open the generated file. Show 8-page official document with headers, radar charts, standards tables, and sign-offs.	"With one click, the officer generates an 8-page official compliance dossier with pre-drafted GFR tender clauses ready for GeM or CPPP bidding."	8-Page Structured PDF (No Screenshot Clipping)
3:30 - 4:15	Switch Persona to Priya Sharma (Govt Dept). Show macro-compliance & road/water tenders. Switch Persona to Amit Verma (PSU Executive). Show 33kV Transformers & high-voltage audits.	"Each role experiences a completely tailored dashboard. Government Departments see state-wide compliance; PSUs see heavy high-voltage equipment verification."	True Role Segregation (Not Cosmetic UI)
4:15 - 5:00	Switch to Admin (Anveshak Lead Administrator). Open Standards Intelligence Registry (/admin/standards). Click "Sync Standards" (show terminal log). Show "New Revision Detected" (IS 1234:2022 vs 2026). Click "Approve & Publish". Open Audit Logs.	"Standards evolve. Our Ingestion Pipeline checks gazette feeds, detects revisions, and allows administrators to publish updates into the live engine with 1 click."	Standards Ingestion & Audit Trail Governance

06

Continuous Standards Ingestion & Lifecycle Pipeline

Judges frequently ask: "What happens when a standard changes or an amendment is published?" Anveshak features an authoritative Standards Ingestion Pipeline (/admin/standards) that prevents regulatory decay:



07

Statutory Certification & QCO Intelligence Layer

Every recommended item passes through a dedicated statutory conformity assessment gate that prevents illegal procurement of uncertified goods:

- **1. Conformity Assessment Scheme:** Distinguishes between BIS ISI Product Certification (Scheme-I), MeitY Compulsory Registration Scheme (CRS Scheme-II), and BEE Star Labeling.
- **2. Statutory Mandate Classification:** Labels requirements as Mandatory (backed by an enforceable Gazette QCO) or Recommended (quality quality best practice).
- **3. Regulatory Provenance:** Cites the notifying ministry (DPIIT, Ministry of Steel, MeitY, MoHUA), gazette order reference number, and effective effective date.
- **4. Mandatory Indenting Action:** Explicitly instructs officers: "Bidders must hold active BIS License (CML / R-Number). Verify license validity on validity on e-BIS portal (manakonline.in) prior to financial bid opening."

Demonstration Anchor for Admin Review

During the live demo, show the pending revision card for IS 1234:2026. Click "Approve & Publish Revision" in front of the judges. Explain that the active registry immediately promotes the new standard and supersedes IS 1234:2022, proving that Anveshak is an evolving regulatory system, not a static database.

08

The 4-Layer AI Hallucination Defense Matrix

In public procurement, hallucinating an incorrect standard can cause contractor lawsuits or dangerous structural failures. Anveshak eliminates hallucination risks via four architectural constraints:

- **1. Closed-World Retrieval Boundary:** The AI model cannot generate arbitrary standards out of its generative memory. Candidate standards standards must exist in our indexed Bureau of Indian Standards knowledge base.
- **2. Mandatory Verbatim Grounded Quotation:** Every recommended standard must cite an exact sentence from the uploaded tender document document justifying the match.
- **3. Uncertainty Thresholding (<70%):** If match confidence drops below 70%, the system flags a prominent amber "Manual Review Recommended" warning badge rather than guessing.
- **4. Statutory Legal Disclaimers:** Every screen and generated PDF explicitly notes that Anveshak acts as decision support under GFR 2017, 2017, and final verification must occur on manakonline.in.

Technology Stack Actually Used & Architectural Justification

When judges ask why specific technologies were selected, provide these crisp, engineering-focused justifications:

TECHNOLOGY LAYER	EXACT TOOL USED	WHY THIS TECHNOLOGY WAS CHOSEN OVER ALTERNATIVES
Frontend Framework	React 18.3	Component reusability, virtual DOM diffing performance, mature ecosystem for PDF/speech integration, and seamless state synchronization.
Tooling & Bundler	Vite 8.2	Sub-500ms production builds (469ms), instant HMR, optimized tree-shaking, and lightweight minified bundles (295kB gzipped).
Styling Architecture	Tailwind CSS v3	Zero-runtime utility CSS, custom Indian Government color tokens (gov-*), responsive mobile/desktop layouts, and accessible contrast.
UI Iconography	Lucide React	Consistent, lightweight SVG vector icons across dashboards, status badges, and action tables.
Speech-to-Text	Web Speech API	Client-side neural speech recognition for English (en-IN) and Hindi (hi-IN) without latency or external cloud API costs.
PDF Generation Engine	jsPDF v4	Client-side programmatic vector PDF generation, creating multi-page structured compliance dossiers without headless browser overhead.
Backend Runtime	Node.js (ESM)	Event-driven, non-blocking asynchronous I/O ideal for handling concurrent multi-page tender document uploads.
REST Framework	Express.js v4	Lightweight, battle-tested HTTP routing, authentication middleware, and robust error handling.
Auth & Security	JWT + bcryptjs	Signed HMAC-SHA256 tokens with role-based claims and salted password hashing (salt rounds: 10).
Database	MongoDB + Mongoose	Flexible document model for complex hierarchical procurement analyses, normative references, and standards catalogs.
Resilience Layer	Dual In-Memory Store	Complete deterministic in-memory fallback layer ensuring 100% offline and serverless resilience during hackathon judging.
Cloud Hosting	Vercel Serverless	Global edge CDN deployment with automatic CI/CD git integration and sub-second serverless response times.

System Strengths vs. Honest Limitations

- **Core Strength 1: Grounded Explainability:** Every recommendation cites verbatim tender text and explains the specific engineering rationale, rationale, eliminating black-box mistrust.
- **Core Strength 2: Dual Resilience Guarantee:** If MongoDB disconnects on serverless, the in-memory fallback engages automatically with 42 with 42 realistic analyses and 23 standards.
- **Honest Limitation 1: Simulated External Feed:** Because BIS does not offer an authenticated public REST API, our Ingestion Pipeline runs runs against a deterministic simulated feed. The update engine itself is 100% functional.
- **Honest Limitation 2: Advisory Scope:** Anveshak provides decision support; statutory responsibility under GFR 2017 remains with the procuring officer.

Top 15 Likely Judge Questions & Winning Answers (Part 1)

Memorize these concise, authoritative model answers. They cover the toughest conceptual, AI, and regulatory questions:

Q1: Why can't officers just search Google or the BIS website?

Google and the BIS portal perform basic keyword searches. But Indian public procurement is governed by multi-standard dependency graphs. For example, procuring an LED street light requires IS 10322 for luminaires, IS 15885 for driver safety, IS 16107 for photometry, and DPIIT Quality Control Orders for mandatory certification. Keyword search cannot extract 12 engineering parameters from a 50-page PDF tender, cannot detect missing 10kV surge protection clauses, and cannot generate a GFR-compliant tender schedule.

Q2: Where exactly is AI used vs. standard database queries?

AI is used in three specific layers: 1) Document Intelligence: Parsing unstructured tender PDFs, tokenizing clauses, and extracting 12 technical engineering parameters. 2) Bilingual Speech-to-Text: Native neural speech recognition for English and Hindi. 3) Semantic RAG Retrieval: Mapping natural language procurement intent to standard scopes using vector similarity and normative graph traversal. Database queries are only used after AI retrieval to enforce deterministic QCO mandates and fetch metadata.

Q3: How do you prevent the AI from hallucinating a non-existent Indian Standard?

We enforce Grounded RAG with a closed-world retrieval boundary. The AI model is strictly prohibited from generating standard strings from its generative memory. Every recommended standard must exist in our indexed Bureau of Indian Standards database, and every match must extract and cite verbatim text from the user's uploaded document.

Q4: What happens if an officer enters an ambiguous or incomplete specification?

Anveshak calculates a Procurement Readiness Score (0–100). If an officer enters a vague specification like "water pump", the completeness score drops below 70%, triggering a prominent amber "Manual Review Recommended" warning badge. The system prompts the officer with clarifying questions (head, flow rate, motor rating) rather than making a wild guess.

Q5: How does the system handle newly published standards or amended regulations?

Anveshak features an administrative Standards Ingestion Pipeline (/admin/standards). The engine checks gazetted feeds, detects version changes (e.g. comparing IS 1234:2022 against newly published IS 1234:2026), and places them in a review queue. Administrators inspect side-by-side version diffs and click "Approve & Publish Revision", which immediately updates the live recommendation engine without requiring a system restart.

Q6: How is the Procurement Readiness Score calculated?

The Readiness Score is a multi-pillar index evaluated from 0 to 100: Technical Parameter Completeness (35 pts: evaluates whether power, material, rating, and ingress are defined), Normative Testing Rigor (25 pts: verifies physical and electrical testing standards), Statutory QCO Coverage (25 pts: confirms mandatory licensing schemes), and Ambiguity Risk (15 pts: penalizes conflicting requirements).

Q7: Is the Role-Based Access Control real or just UI cosmetics?

It is strictly enforced server-side. Authentication generates signed HMAC-SHA256 JWT tokens containing role claims. Backend Express routes enforce authorizeRoles("admin", "Organization/Admin"). If a Procurement Officer tries to access /api/admin/standards or /api/admin/users, the server returns an immediate HTTP 403 Forbidden. Furthermore, dashboards, metrics, and report archives are strictly filtered by user role and organization.



Top 15 Likely Judge Questions & Winning Answers (Part 2)

Q8: How does Anveshak support Hindi and regional language procurement officers?

Anveshak includes a full bilingual architecture: 1) Speech-to-Text with native Hindi (hi-IN) neural recognition and live transcript preview. 2) Natural language understanding that accepts Hindi and Hinglish inputs (e.g. "8!< G 2? 100W LED 2> "). 3) A global Language dashboard statistics, status badges, and action buttons into Hindi.

Q9: How do you handle complex scanned PDFs with poor print quality?

Anveshak uses an optical text layout segmenter. If a document has low text density, the system extracts available text snippets, flags missing parameters, and asks the officer to verify critical values. In our product roadmap, we are integrating specialized neural document layout models (PaddleOCR / LayoutLM) for degraded photocopies.

Q10: Can a vendor or contractor game the system by using misleading tender keywords?

No, because Anveshak does not rely on single keyword matching. The engine performs multi-factor parameter validation. If a vendor advertises a consumer-grade LED panel as a "high-bay municipal street light", Anveshak flags deviations in thermal dissipation, IP ingress protection (IS/IEC 60529), and driver surge immunity (IS 16107), exposing the non-compliance.

Q11: What is the legal validity of Anveshak's recommendations under GFR 2017?

Anveshak acts as an intelligent decision-support copilot under Rule 144 of GFR 2017. It empowers indenting officers to draft legally compliant specifications and pre-qualifying conditions. However, the final statutory responsibility remains with the procuring officer, which is why all generated reports feature formal disclaimers and verification links to manakonline.in.

Q12: How does the system scale if adopted nationally across thousands of government offices?

The application is architected as stateless microservices deployed on serverless edge functions. Database lookups are indexed across standard numbers, categories, and organizations. The document processing pipeline is decoupled, allowing asynchronous worker queues (BullMQ / Redis) to process multi-hundred-page tender archives concurrently without blocking API responsiveness.

Q13: How do you ensure enterprise data privacy and security for sensitive tenders?

All communications use HTTPS/TLS. Passwords are salted and hashed using bcryptjs. All API calls require signed HMAC-SHA256 JWT tokens with 30-day expiration. Furthermore, all user actions (tenders uploaded, reports generated, roles modified, standards synced) are permanently recorded in an immutable System Audit Trail (/admin/audit-logs) with timestamps, user IDs, and IP addresses.

Q14: How does Anveshak integrate with the Government e-Marketplace (GeM)?

Anveshak generates GFR-compliant 8-clause tender schedules and structured PDF dossiers formatted specifically to match GeM's Custom BOQ and Technical Specification bidding categories. In our future roadmap, we have architected REST API webhooks to integrate directly into GeM's tender creation workflow.

Q15: What makes Anveshak commercially viable and adoptable by government departments?

Procurement errors cause billions of rupees in arbitration disputes, contract re-tendering delays, and sub-standard infrastructure. Anveshak reduces tender specification drafting time from days to seconds, guarantees 100% QCO compliance, and prevents vendor disputes, delivering immediate Return on Investment (ROI) for any procuring department.

Critical Delivery Rules: What NOT to Say in Front of Judges

Experienced hackathon judges look for over-claiming and fragile architecture. Protect your presentation by following these communication rules:

DON'T SAY:

"Our AI knows all 21,000 standards and never makes mistakes."

INSTEAD SAY:

"Our engine uses Grounded RAG with closed-world verification. It only retrieves verified standards and flags uncertainty below 70%."

DON'T SAY:

"We built our own custom LLM from scratch."

INSTEAD SAY:

"We built a domain-specific hybrid architecture combining neural parameter extraction with deterministic standards dependency graphs."

DON'T SAY:

"This replaces the procurement officer completely."

INSTEAD SAY:

"This is an intelligent decision-support copilot that saves days of drafting time while keeping statutory accountability with the officer."

DON'T SAY:

"We are directly connected to live BIS servers today."

INSTEAD SAY:

"Our Ingestion Pipeline is architecturally complete with version diffing and 1-click publishing, currently operating on a deterministic feed."

SIH 2026 Evaluation Rubric Alignment & Final Delivery Checklist

- **1. Innovation & Novelty (20%):** Emphasize the continuous standards ingestion pipeline, grounded evidence citation, and multi-pillar readiness readiness index.
- **2. Technical Complexity (25%):** Highlight the bilingual speech recognition, PDF text segmentation, hybrid RAG scoring, and dual-mode dual-mode database resilience.
- **3. Real-World Feasibility (25%):** Stress alignment with GFR 2017 Rule 144 and DPIIT Quality Control Orders public expenditure.
- **4. UI/UX Polish & Accessibility (15%):** Point out the clean institutional aesthetics, responsive navigation, and bilingual English/Hindi interface. interface.
- **5. Presentation & Defense (15%):** Deliver crisp, confident answers to the Top 15 Judge Questions without hesitation.

Team Presentation Division of Roles

- **Speaker 1 (The Hook):** Explains the 150 Lakh Crore problem, GFR Rule 144, and delivers the 2-minute
- **Speaker 2 (The Demonstration):** Controls the screen, runs the 5-minute live demo, and switches roles.
- **Speaker 3 (The Defense):** Explains the technical architecture, AI hallucination strategy, and leads the judge Q&A.